

Dietary Counseling for High Blood Cholesterol in Families at Risk of Coronary Disease

Taryn F. Moy, MS, RD; Lisa R. Yanek, MPH; Judith V. Raqueño, BSN; Pamela J. Bezirdjian, MPH; Roger S. Blumenthal, MD; Lora B. Wilder, ScD, RD; Diane M. Becker, ScD, MPH

A positive family history of coronary heart disease alone confers an increased risk, which may be affected by untreated hypercholesterolemia. Dietary counseling is a first-line treatment approach. To determine whether nurse counseling can provide additional benefits over usual physician efforts to lower dietary fat in high-risk persons, 117 apparently healthy adult siblings of persons with premature coronary heart disease were counseled by a registered nurse using adapted national guidelines. Reductions in total fat, saturated fat, and cholesterol were significantly greater in the nurse group compared to those in the usual care group. Total fat intake decreased by 14 g in the nurse group, compared with an increase of 5 g in the usual care group ($p=0.0001$). Assignment to the nurse group was also a significant predictor of a greater reduction in the percentage of total fat calories ($p=0.008$). The authors conclude that a registered nurse may serve as a complement to usual care in efforts to lower dietary fat and cholesterol in high-risk families. (Prev Cardiol. 2001;4:158-164) ©2001 CHF, Inc.

Healthy brothers and sisters of individuals with premature coronary heart disease (CHD) are at a markedly increased risk for developing premature CHD themselves.¹ Clustering of highly prevalent and modifiable CHD risk factors has also been shown in these brothers and sisters,²⁻³ as have unhealthy eating behaviors.⁴

The National Cholesterol Education Program Adult Treatment Panel II (ATP II) guidelines⁵ recommend dietary intervention as the first-line approach

in the treatment of hypercholesterolemia. The new ATP III guidelines will probably again place emphasis on the use of dietary interventions as an adjunct to pharmacologic treatment. Physicians are encouraged to provide initial dietary counseling. The guidelines suggest that registered dietitians and qualified nutritionists serve as consultants if initial attempts by primary care providers to alter diet are unsuccessful. The relative effectiveness of dietary counseling by dietitians, nurses, and physicians remains unclear.⁶⁻⁹ Studies have demonstrated that only a minority of hypercholesterolemic patients are referred to dietitians by physicians and that most physicians are unable to provide intensive dietary counseling themselves.¹⁰⁻¹³ Given that the number of available dietitians is small relative to that of nurses, and that physicians alone may not be able to concentrate on dietary counseling in clinical encounters, the purpose of this study was to determine whether a registered nurse, trained in dietary intervention, could effectively counsel high-risk individuals to lower their total fat and cholesterol intake. This is important both because of the primary impact of diet and because of the role diet may play in augmenting the efficacy of the statin drugs. In an era when drugs are being increasingly used, effective ways to implement dietary intervention remain important. Thus, we compared nurse counseling with usual care by a physician in healthy, free-living brothers and sisters of persons with premature CHD. This was part of a larger prospective study of coronary disease risk factors in persons with familial-aggregated premature CHD.

METHODS

Study Design

From 1991-1994, 327 apparently healthy 30-59-year-old brothers and sisters of patients with documented CHD diagnosed before 60 years of age were recruited for a full-day cardiovascular screening to assess risk factors. Eligible siblings met study entry criteria by having at least one of the following risk factors: low-density lipoprotein cholesterol (LDL-C)

From The Center for Health Promotion and Division of Cardiology, The Johns Hopkins Medical Institutions, Baltimore, MD

Address for correspondence/reprint requests:

Taryn F. Moy, MS, RD, Research Associate, Medicine, 1830 East Monument Street, Room 8024, Baltimore, MD 21205

Manuscript received June 23, 2000; accepted July 6, 2000

≥ 3.4 mmol/L (130 mg/dL), blood pressure $\geq 140/90$ mm Hg or current use of antihypertensive medication, or current smoking. Of the 235 enrolled, 117 siblings were randomly assigned to nurse counseling and 118 to usual care by their primary physicians. Randomization was by family, via a computerized schema, after all eligible siblings from the same family had been screened. ATP II dietary guidelines were applied for 2 years. Participants were either seen by the nurse or, in the case of the usual care group, were referred to their physician providers for dietary management.

Results and feedback from the baseline screening, with patient-specific recommendations based on national guidelines, were sent to all participating siblings and all primary care physicians in both groups. Physicians in the usual care group received specific, detailed recommendations for risk factor management on three occasions (baseline, 1- and 2-year follow-up). Physicians of siblings in the nursing group were asked explicitly not to manage dietary interventions. Follow-up at 1 and 2 years was conducted in both groups. Details regarding overall study design of the larger prospective cohort study are described in a prior publication.⁴ This paper reports only on the results from the dietary intervention. The study was approved by the Johns Hopkins Joint Committee on Clinical Investigation. All participants gave informed consent.

Baseline Measurement

Height and weight were measured with the subjects in light indoor clothing. Body mass index was calculated as weight in kilograms divided by height in meters squared. An average of three auscultatory blood pressure measurements were taken, according to American Heart Association guidelines.¹⁴

Blood specimens were obtained by venipuncture after participants had fasted for 12 hours. Plasma total cholesterol, triglycerides (TG), and high-density lipoprotein cholesterol levels were measured directly in the Johns Hopkins Chemistry Laboratory, by Centers for Disease Control and Prevention standardized methods.¹⁵ LDL-C was calculated for individuals with TG levels < 4.5 mmol/L (400 mg/dL), using the Friedewald formula.¹⁶ Individuals with TG levels greater than this ($n=21$) were excluded from lipid analyses because of an inability to evaluate LDL-C changes.

Dietary Assessment

Dietary information was assessed at baseline and 2-year follow-up, using a semi-quantitative food frequency questionnaire, the Health Habits and History Questionnaire developed by Block et al.¹⁷ The instrument was modified to reflect dietary intake over the previous month, instead of the previous year, as had been the convention, because our prior research in this population and data from other studies suggested that short-term memory is better. Reference to long-term

periods, such as "in the past year" have not been found to provide better information about "usual" diet compared with shorter reference periods.¹⁸ In fact, the accuracy of episodic memory of diet worsens as the length of time increases.¹⁸ Participants were asked about their usual eating habits "over the last month" vs. "over the last year." Nutrient analysis was determined by the Dietary Analysis System for personal computers.¹⁹ Nutrient estimates were based on the National Health and Nutrition Examination Survey II (NHANES II) nutrient data base. In published studies, validation of the original Health Habits and History Questionnaire revealed correlations for nutrients between the instrument and multiple-day dietary records that ranged between 0.4 and 0.7.²¹⁻²³ Test-retest reliability coefficients ranged between 0.5 and 0.8.²⁴

Questionnaires in both groups were self-administered and were reviewed with the participants for completeness and accuracy. All dietary questionnaires were reviewed by the study dietitian.

Nurse Training

There were three nurses over the course of the study period. Each nurse was trained didactically and experientially, and observed multiple individual counseling sessions with dietitians with expertise in lipid management. Training in the ATP II guidelines and behavioral modification strategies by the principal investigator and the dietitian averaged 40 hours for the nurse counselors. The nurse functioned independently as a dietary counselor, although the study dietitian provided consultation for a few participants who had specific eating problems (i.e., multiple food allergies).

Nutrition Intervention

All nurse-counseled participants received individualized instructions to lower total fat intake, based on the ATP II dietary guidelines. Interventions focused primarily on total fat consumption and daily monitoring. This modification was based on preliminary work in which reductions in saturated fat and cholesterol to goal levels were observed in virtually all participants who reduced total fat intake (changes in total fat and saturated fat were correlated: $r=0.98$; $p=0.0001$). Participants were initially given a total "fat allowance," based on their intake assessed at baseline, their current lifestyles, and stated willingness to change. This was usually a goal of < 40 g of total fat. They were then taught how to read food labels and use a "fat counter" (a pocket-sized booklet) to monitor and record total daily fat intake. Participants negotiated their fat gram intake up or down with the nurse in counseling sessions. The self-monitoring logs were used for the first 1-2 months, although fewer than one half of the group complied regularly. After that, some individuals chose to continue using them, and others did not. Most subjects became very aware of their usual daily fat gram intake by using the fat counter and did not record their data but were able to recount it.

Table I. Demographic and Risk Factor Characteristics of Siblings by Group at Baseline

	NURSE COUNSELING (N=117)	USUAL CARE (N=118)	<i>p</i> *
Age (means±SD)	46.2±7	45.7±7	0.597
Education (means±SD)	12.3±3	12.6±2	0.434
Sex (% male)	55	49	0.395
Race (% African American)	18	25	0.165
Married (%)	75	73	0.644
Body mass index (kg/m ²) (means±SD)	28.5±5	29.5±7	0.181
LDL-C (mmol/L) (means±SD) (mg/dL)	4.7±1.4 (181.9±54.2)	4.3±1.2 (166.4±46.4)	0.037
Obesity** (%)	30	36	0.354
High LDL-C† (%)	84	75	0.084
Hypertension†† (%)	57	56	0.837
Current smoking (%)	38	53	0.015
High LDL-C† and hypertension†† (%)	37	27	0.113
High LDL-C† and current smoking (%)	15	20	0.322
Hypertension†† and current smoking (%)	2	6	0.092
High LDL-C,† current smoking, and hypertension†† (%)	13	15	0.591

LDL-C=low-density lipoprotein cholesterol;
 *Between-group significance: Student *t* test for continuous variables and contingency table arrays with the χ^2 statistic for categoric variables; **body mass index ≥ 30 kg/m²; †LDL-C ≥ 3.4 mmol/L (130 mg/dL); ††blood pressure $\geq 140/90$ mm Hg and/or current antihypertensive medication

Participants were seen individually and with family members, approximately every 6–8 weeks to reinforce the diet, evaluate dietary compliance, and measure lipids. Visit intervals were not rigid and there was considerable variation in their frequency, which was negotiated between the nurse and participant. Some encounters occurred over the telephone. At each visit, a standardized, brief dietary fat screening instrument developed by Block et al.²⁵ was utilized to identify potential problems and provide a springboard for discussion of eating habits. Counseling was individualized according to the participants' initial dietary habits and lifestyles, as well as their progress at each visit, to allow the same flexibility used in standard, usual care medical practice.

Statistical Analyses

Statistical analyses were performed with SAS (Cary, NC).²⁶ Differences from baseline to the 2-year follow-up between and within the nurse and usual care groups were examined by Student's *t* test for continuous variables, and contingency table arrays and the χ^2 statistic were applied to categoric variables. All analyses were based on "intention to treat." Baseline values were used for all siblings for whom data were incomplete at 2-year follow-up (*n*=55). These individuals were also assigned a zero for all change scores. Overall return rates were 73% for the nurse group and 81% for the usual care group.

Multiple linear regression was performed to predict the change in percentage of calories from total fat. Achievement of the goal level of ≤ 40 g fat/day was assessed by multiple logistic regression.

The Generalized Estimating Equations method²⁷ was applied to regression analyses to control for any possible intrafamilial clustering, although none was observed on examination.

RESULTS

Baseline characteristics are presented in Tables I and II. Dietary intake of total fat, saturated fat, cholesterol, and calories in the two groups were similar. Over 80% of both groups consumed more total fat than the nationally recommended goal of $\leq 30\%$ of total calories. The usual care group had a significantly lower mean LDL-C and a significantly higher prevalence of smoking as compared to the nursing group at baseline. The mean body mass index shows that overweight was prevalent in both groups. The distributions of siblings in each group with various combinations of risk factors are also presented in Table I.

Table III presents within- and between-group changes from baseline to 2-year follow-up. Thirty percent of the nurse group and 17% of the usual care group decreased their mean percent fat intake to the national guideline of $\leq 30\%$ (*p*=0.019). The nurse group decreased their mean total fat intake to about 34%, whereas in the usual care group an average of approximately 38% of calories were derived from total fat at follow-up. Body mass index decreased slightly in the nurse group and increased in the usual care group, although these changes were not significant. The mean decrease in LDL-C was significant in the nurse group, and there was a trend toward favorable changes in high-density lipoprotein and TG.

Table II. Baseline Characteristics of Dietary Factors in Nurse Counseling and Usual Care Groups

	NURSE COUNSELING (N=117) MEANS $\pm$ SD	USUAL CARE (N=118) MEANS $\pm$ SD	<i>p</i> *
Total fat (g)	85.1 $\pm$ 42	85.0 $\pm$ 40	0.974
Saturated fat (g)	30.2 $\pm$ 16	29.7 $\pm$ 15	0.793
Monounsaturated fat (g)	32.1 $\pm$ 17	31.6 $\pm$ 15	0.811
Polyunsaturated fat (g)	16.0 $\pm$ 8	16.9 $\pm$ 10	0.449
Cholesterol (mg)	299.8 $\pm$ 168	291.8 $\pm$ 140	0.691
Kilocalories	1977 $\pm$ 777	1978 $\pm$ 752	0.994
% kcal from total fat	38.0 $\pm$ 7	38.3 $\pm$ 8	0.785
% kcal from fat $\leq$ 30%	13	14	0.867

*Between-group significance: Student *t* test for continuous variables and contingency table arrays with the χ^2 statistic for categorical variables

Linear regression analysis, including age, sex, race, baseline body mass index, baseline LDL-C, and intervention group assignment, was performed to predict the change in percent of calories from total fat at the end of the 2-year study (Table IV). Assignment to the nurse group was significantly predictive of a greater change in the percentage of fat caloric intake ($p=0.001$).

In multiple logistic regression analysis, controlling for sex, race, body mass index at baseline, and LDL-C at baseline, the nurse group was statistically significantly more likely than the usual care physician group to achieve the goal of ≤ 40 g of total fat per day (odds ratio, 2.50; 95% confidence interval, 1.1–6.0).

DISCUSSION

Physician training in nutrition has been noted as inadequate.^{28,29} The consistent inability of usual medical care systems to provide adequate counseling^{9,30} provided a powerful impetus to develop a pragmatic model of nutrition counseling to be offered in clinical settings. In this study, a flexible nurse-mediated counseling intervention based on national guidelines appeared to be successful in favorably altering diet in an often untreated, exceedingly high-risk population. While the usual physician care group also showed some improvement, the nurse group made more significant progress toward achieving the cholesterol goals. National surveys have found that adults in the general population derive about 33%–36% of their energy from fat.^{31,32} The high-risk siblings in this study acquired about 38% of their diet as fat at baseline. While one third of adults reportedly consume $\leq 30\%$ of calories from fat,³¹ only 13% of our nurse group and 14% of our usual care group met the national goal level at baseline—a rather alarming number, given their high-risk status.

There have been very few other studies showing that dietary counseling for hyperlipidemia by nondietitians, and specifically by nurses, is effective. Blair et al.³³ showed that a cardiology clinic nurse could suc-

cessfully lower lipid levels in 37% of patients with diet alone ($n=32$), and in 63% of patients ($n=54$) with a combination of diet and cholesterol-lowering drugs. However, this was an earlier study in which the emphasis was on dietary cholesterol vs. dietary fat, and the method of dietary assessment was not described.

Roderick et al.³⁴ studied the effectiveness of specially trained nurses providing subjects with standard written health education materials and compared it with usual care practices in England and Wales. They observed modest but significant changes in diet and associated risk factors in the nurse group and concluded that the intervention could be beneficial to those at high risk for CHD.³⁴ In our study of high-risk siblings, a significantly greater LDL-C change was observed in the nurse group, which is concordant with the changes in diet that were observed.³⁵ This supports the contention that application of additional resource personnel may be beneficial in high-risk groups.

The US Department of Agriculture Dietary Guidelines for Americans,³⁶ the American Heart Association,³⁷ and the US Public Health Service³⁸ recommend that all primary care providers offer their patients counseling to promote a healthy diet, increased physical activity, and smoking cessation as part of the preventive health examination. It is estimated that approximately 52 million Americans should receive dietary education, according to the ATP II recommendations.³⁹ However, physicians often do not counsel or refer patients to dietitians,^{9,11,28,40} and when they do, contact with patients is often brief and limited to a single visit.^{10,30} It appears to be the norm that physicians have insufficient time to spend on nutrition counseling. Ockene et al.³⁰ reported that the average time spent for nutrition counseling in usual care was only 2.7 minutes, and Kushner²⁸ reported that the majority of physicians spend 5 minutes or less. This brief encounter would not reasonably be expected to produce significant and sustained changes in eating behavior.⁴¹ In our study, encounters lasted an aver-

Table III. Changes in Diet, Lipids, and Body Mass Index by Group From Baseline to 2 Years

	NURSE COUNSELING (N=117) MEANS±SD	USUAL CARE (N=118) MEANS±SD	<i>p</i> *
Total fat (g)	-14.3±34	4.7±41	0.0001
<i>p</i> **	0.0001	0.2125	
Saturated fat (g)	-4.9±12	1.9±16	0.0002
<i>p</i> **	0.0001	0.1875	
Cholesterol (mg)	-27.3±122	19.6±97	0.0013
<i>p</i> **	0.0173	0.0303	
Kilocalories	-152.3±616	114.0±704	0.0023
<i>p</i> **	0.0085	0.0811	
% Kcal from total fat	-3.9±8	-0.27±7	0.0001
<i>p</i> **	0.0001	0.6623	
% Kcal from saturated fat	-1.4±3	-0.0064±3	0.0005
<i>p</i> **	0.0001	0.9809	
Body mass index (kg/m ²)	-0.10±1	0.21±2	0.164
<i>p</i> **	0.4398	0.2469	
LDL-C (mmol/L)	-0.69±1.1	-0.4±0.8	0.0223
(mg/dL)	(-26.7±42.6)	(-15.5±31.0)	
<i>p</i> **	0.0001	0.0001	
HDL-C (mmol/L)	0.044±0.3	0.008±0.2	0.240
(mg/dL)	(1.7±11.6)	(0.3±7.7)	
<i>p</i> **	0.0621	0.6778	
Triglycerides (mmol/L)	-0.4±2.0	-0.06±1.9	0.183
(mg/dL)	(-35.4±177.2)	(-5.3±168.3)	
<i>p</i> **	0.0334	0.7063	

LDL-C=low-density lipoprotein cholesterol; HDL-C=high-density lipoprotein cholesterol;

*Between-group significance; **within-group significance

age of 2 hours at the first visit, and 45–60 minutes on subsequent visits. Telephone counseling was usually participant-initiated and the sessions averaged 20 minutes each. Although the minimum frequency of visits was determined by national guidelines, participants were able to visit more frequently if they so desired. On average, there were 6.5 nurse encounters per year, including in-person and telephone sessions. This is more than a usual care physician might be able to manage.

There are also other barriers to physicians delivering dietary interventions. Among those identified in prior studies are: inadequate training in nutrition counseling, lack of behavioral intervention skills, lack of reimbursement mechanisms, a belief that physicians are ineffective in dietary intervention, and a belief that chronic disease risk factor interventions are of low priority in acute care practice settings.^{12,28,29} Although these factors may be changing, there have not been dramatic alterations in medical practice since publication of the ATP II guidelines. With the release of the new ATP III guidelines, the crucial need for implementing dietary plans will be more evident.

Health manpower estimates suggest that the demand for available dietitians may exceed the supply.⁴² The shift to managed care is also transforming the role of some dietitians from direct providers to facilitators

of nutrition services. Rather than managing patients, dietitians are often utilized to develop and implement programs, train personnel, and monitor aggregate outcomes in a practice setting.⁴³ Public health nutritionists are also in short supply,⁴⁴ and most are committed to maternal and child health services and therefore are unavailable for chronic disease prevention interventions.⁴⁵ The need for other health care providers to extend the services of nutritionists is crucial.

In conclusion, nurses may have opportunities to reach a large number of patients for both initial dietary intervention and regular follow-up. The number of nurses in the health workforce far exceeds that of dietitians⁴² and there is a general shift in nursing practice from hospital acute care to outpatient care settings, most often in physician practices. Physicians in clinical practice, where high-risk families can be identified, may successfully augment their nutritional counseling services through nurses trained in dietary counseling.

This study demonstrates that effective counseling by trained nurses can lead to beneficial long-term dietary change in a high-risk population that is easily identifiable at the time of a premature CHD event. Given the release of the new ATP III guidelines, application of this model could potentially impact millions of individuals identified as high-risk and in need of dietary intervention for hypercholesterolemia. The flexi-

Table IV. Multiple Linear Regression: Predictors of Change in Dietary Fat at 2 Years

CHANGE IN % KILOCALORIES FROM TOTAL FAT*		
	PARAMETER ESTIMATE	P
Age (years)	-0.0413	0.519
Sex (female)	1.14	0.220
Race (African Americans)	2.00	0.084
Body mass index (kg/m ²)	0.0975	0.238
Baseline LDL-C	-0.0067	0.468
Intervention group (nurse counseling)	-3.27	0.0011

LDL-C=low-density lipoprotein cholesterol;
 *Change in % kilocalories from total fat=baseline % calories from total fat minus % calories from total fat at 2 years

bility of the approach and the latitude the nurses have in making decisions that integrate cholesterol lowering with participants' social and personal needs make this model particularly relevant to real-world practice.

This study was supported by grant NR0224103 from the National Institute of Nursing Research. Computational assistance was provided by General Clinical Research Center Grants MO1-RR00035 and MO1-RR00722 from the National Institutes of Health, Bethesda, MD.

REFERENCES

- 1 Blumenthal RS, Becker DM, Moy TF, et al. Exercise thallium tomography predicts future coronary heart disease events in a high risk asymptomatic population. *Circulation*. 1996;93:915-923.
- 2 Genest J, Cohn JS. Clustering of cardiovascular risk factors: targeting high-risk individuals. *Am J Cardiol*. 1995;76:8A-20A.
- 3 Sacco RL, Kargman DE, Gu Q, et al. Race-ethnicity and determinants of intracranial atherosclerotic cerebral infarction: the Northern Manhattan Stroke Study. *Stroke*. 1995;26:14-20.
- 4 Allen JK, Young DR, Blumenthal RS, et al. Prevalence of hypercholesterolemia among siblings of persons with premature coronary heart disease: application of the Second Adult Treatment Panel Guidelines. *Arch Intern Med*. 1996;156:1654-1660.
- 5 National Cholesterol Education Program. Second report of the expert panel on detection, evaluation, and treatment of high blood cholesterol in adults (Adult Treatment Panel II). *Circulation*. 1994;89:1329-1445.
- 6 McBride PE, Plane MB, Underbakke G. Hypercholesterolemia: the current educational needs of physicians. *Am Heart J*. 1992;123:817-824.
- 7 Scott CS, Leaf D, Neighbor WE. Preventive cardiology education and practice in residency training: residents' attitudes, perceptions, and practices. *Am J Prevent Med*. 1990;6(suppl 2):60-69.
- 8 Rhodes KS, Bookstein LC, Aaronson LS, et al. Intensive nutrition counseling enhances outcomes of National Cholesterol Education Program dietary therapy. *J Am Diet Assoc*. 1996;96:1003-1010.
- 9 Schectman JM. Association between physician counseling for hypercholesterolemia and patient dietary knowledge. *Am J Prevent Med*. 1994;10:136-139.
- 10 Hyman DJ, Clark M, Houston-Miller N, et al. Cholesterol-related counseling by registered dietitians in northern California. *Prevent Med*. 1992;21:746-753.
- 11 O'Keefe CE, Hahn DF, Betts NM. Physicians' perspectives on cholesterol and heart disease. *J Am Diet Assoc*. 1991;91:189-192.
- 12 Ockene IS, Ockene JK. Barriers to lifestyle change, and the need to develop an integrated approach to prevention. *Cardiol Clin*. 1996;14:159-169.
- 13 Schucker B, Wittes JT, Santanello NC, et al. Change in cholesterol awareness and action. Results from national physician and public surveys. *Arch Intern Med*. 1991;151:666-673.
- 14 Frolich ED, Grim C, LaBarthe DR, et al. Recommendations for human blood pressure determination by sphygmomanometers. Report of a special task force appointed by the Steering Committee, American Heart Association. *Hypertension*. 1988;11:209A-222A.
- 15 National Cholesterol Education Program. *Recommendations on lipoprotein measurement*. US Department of Health and Human Services, Public Health Service, National Institutes of Health, National Heart, Lung, and Blood Institute, Bethesda, MD; 1995. Publication No. 95-3044.
- 16 Friedewald WT, Levy RI, Fredrickson DS. Estimation of the concentrations of low-density lipoprotein (LDL) cholesterol in plasma, without use of the preparative ultracentrifuge. *Clin Chem*. 1972;18:499-502.
- 17 Block G, Hartman AM, Dresser CM, et al. A data-based approach to diet questionnaire design and testing. *Am J Epidemiol*. 1986;124:453-469.
- 18 Smith AF, Jobe JB, Mingay DJ. Retrieval from memory of dietary information. *Appl Cognit Psychol*. 1991;5:269-296.
- 19 Smucker R, Block G, Coyle L, et al. A dietary and risk factor questionnaire and analysis system for personal computers. *Am J Epidemiol*. 1989;129:445-449.
- 20 NHANES II. National Health and Examination Survey. Wyattsville, MD: National Center for Health Statistics; 1976-1980.
- 21 Block G, Woods M, Potosky A, et al. Validation of a self-administered diet history questionnaire using multiple diet records. *J Clin Epidemiol*. 1990;43:1327-1335.
- 22 Block G, Thompson FE, Hartman AM, et al. Comparison of two dietary questionnaires validated against multiple dietary records collected during a 1-year period. *J Am Diet Assoc*. 1992;92:686-693.
- 23 Mares-Perlman JA, Klein BEK, Klein R, et al. A diet history questionnaire ranks nutrient intakes in middle-aged and older men and women similarly to multiple food records. *J Nutr*. 1993;123:489-501.
- 24 Bittoni MA, Wilkins JRI. Assessment of the reliability of a diet history questionnaire. *Nutr Cancer*. 1994;21:143-155.
- 25 Block G, Clifford C, Naughton MD, et al. A brief dietary screen for high fat intake. *J Nutr Educ*. 1989;21:199-207.
- 26 SAS Institute Inc. SAS/STAT User's Guide, Version 6.11. Cary, NC: SAS Institute Inc; 1995:773-875.
- 27 Zeger SL, Liang K, Albert PS. Models for longitudinal data: a generalized estimating equation approach. *Biometrics*. 1988;44:1049-1060.

- 28 Kushner RF. Barriers to providing nutrition counseling by physicians: a survey of primary care practitioners. *Prevent Med.* 1995;24:546-552.
- 29 Hiddink GJ, Hautvast JG, Woerkum CM, et al. Nutrition guidance by primary-care physicians: perceived barriers and low involvement. *Eur J Clin Nutr.* 1995;49:842-851.
- 30 Ockene IS, Hebert JR, Ockene JK, et al. Effect of physician-delivered nutrition counseling training and an office-support program on saturated fat intake, weight, and serum lipid measurements in a hyperlipidemic population. *Arch Intern Med.* 1999;159:725-731.
- 31 US Department of Agriculture, Agricultural Research Service. Data tables: Results from USDA's 1996-96 Continuing Survey of Food Intakes by Individuals and 1994-96 Diet and Health Knowledge Survey. *On: 1994-96 Continuing Survey of Food Intakes by Individuals and 1994-96 Diet and Health Knowledge Survey CD-ROM, NTIS Accession Number PB98-500457*, 1997:37.
- 32 Norris J, Harnack L, Carmichael S, et al. US trends in nutrient intake: the 1987 and 1992 National Health Interview Surveys. *Am J Public Health.* 1997;87:740-746.
- 33 Blair TP, Bryant FJ, Bocuzzi S. Treatment of hypercholesterolemia by a clinical nurse using a stepped-care protocol in a nonvolunteer population. *Arch Intern Med.* 1988;148:1046-1048.
- 34 Roderick P, Ruddock V, Hunt P, et al. A randomized trial to evaluate the effectiveness of dietary advice by practice nurses in lowering diet-related coronary heart disease risk. *Brit J Gen Pract.* 1997;47:7-11.
- 35 Becker DM, Raqueño JV, Yook RM, et al. Nurse-mediated cholesterol management compared with enhanced primary care in siblings of individuals with premature coronary disease. *Arch Intern Med.* 1998;158:1533-1539.
- 36 USDA and DHHS (US Department of Agriculture and US Department of Health and Human Services). Nutrition and your health: dietary guidelines for Americans, 4th ed. *Home Garden Bull.* 1995;232:27.
- 37 Grundy SM, Balady GJ, Criqui MH, et al. Guide to primary prevention of cardiovascular diseases: a statement for healthcare professionals from the Task Force on Risk Reduction. *Circulation.* 1997;95:2329-2331.
- 38 US Department of Health and Human Services. Public Health Service. Healthy People 2000: Midcourse Review and 1995 Revisions. DHHS Publication. Washington, DC: US Government Printing Office, 1995 (un-numbered).
- 39 Sempos CT, Cleeman JI, Carroll MD, et al. Prevalence of high blood cholesterol among US adults: an update based on guidelines from the second report of the National Cholesterol Education Program Adult Treatment Panel. *JAMA.* 1993;269:3009-3014.
- 40 Missed opportunities in preventive counseling for cardiovascular disease—United States, 1995. *MMWR.* 1998;47:91-95.
- 41 Elson RB, Splett PL, Bostick RM, et al. Dietitian practices for adult outpatients with hypercholesterolemia referred by physicians. *Arch Fam Med.* 1994;3:1073-1080.
- 42 Cohen MS. Labor shortages as America approaches the 21st century. Ann Arbor, MI: University of Michigan Press; 1995:101.
- 43 Laramée SH. Nutrition services in managed care: new paradigms for dietitians. *J Am Diet Assoc.* 1996;96:335-336.
- 44 Haughton B, Story M, Keir B. Profile of public health nutrition personnel: challenges for population/system-focused roles and state-level monitoring. *J Am Diet Assoc.* 1998;98:664-670.
- 45 Keyserling TC, Ammerman AS, Atwood JR, et al. A cholesterol intervention program for public health nurses in the rural Southeast: description of the intervention, study design, and baseline results. *Public Health Nurs.* 1999;16:156-167.